Joel Jacob

Software Engineer

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Software Engineer with 3 years building backend and full-stack systems in C#/.NET and Java, plus React and Python. Built CI/CD pipelines that cut staging deploys to under 15 minutes and automated reporting used across academic departments. Looking for backend or full-stack roles where I can ship and automate.

SKILLS

Languages	Java, Python, C#, C++, JavaScript, SQL
Frameworks	React, PyTest, Selenium, PyTorch, Pandas, NumPy
Software & Tech	Docker, Azure Pipelines, GitHub Actions
Platforms	Windows, Linux, Unix

EXPERIENCE

Programmer Analyst — Niagara College (ESS)

Welland, ON · Jul 2023 – Jun 2026

- Developed, debugged, and maintained automated reporting scripts for 4 departments (ITS, Enrolment, Finance, HR), cutting a manual reporting cycle of nearly 2 weeks down to a few minutes.
- Architected a standardized CI/CD pipeline for the test environment, replacing a manual email-based approval chain that took 30 minutes to 2 days with one-click approvals, cutting staging deployment times to under 15 minutes.
- Collaborated across business and tech teams to gather requirements and define clear acceptance criteria, consistently closing 20% more tickets per sprint than the team average.

Junior Developer (Co-op) — Digital Garage, MTO

St. Catharines, ON · May 2022 – Aug 2022

- Delivered full-stack features for an internal web application using C#, React, and SQL in an Agile team; refactored repeated logic into shared functions and split overbloated functions into focused components to improve maintainability across the site.
- Prototyped an automated Azure-based virus-scanning API for the data-ingestion pipeline, mitigating vulnerabilities from uploaded documents.
- Built a decentralized application prototype using Solidity and Truffle, evaluating blockchain ledger immutability for task-state management.
- Researched QA testing by developing automated test scripts in Katalon.

Junior Technical Analyst (Co-op) — MTO

St. Catharines, ON · Jun 2020 – Aug 2020

- Optimized Azure cloud governance by analyzing, auditing, and restructuring tagging schemas across resource groups to map asset utilization.
- Facilitated cross-functional meetings to present technical findings, securing stakeholder consensus to decommission redundant cloud resources.
- Audited and eliminated unneeded infrastructure, reducing Azure subscription costs by 38% in the primary environment and 18% in the secondary.

Research Assistant — Brock University

St. Catharines, ON · May 2020 – Jun 2020

- Conducted root-cause analysis on server bottlenecks, evaluating machine-learning methods to identify critical-path anomalies and pinpoint underperformance.
- Independently studied predictive modeling and data-analysis techniques through academic literature, translating algorithms into diagnostic frameworks.

ACADEMIC PROJECTS

Capstone — CI/CD Microservice Deployment (Apr 2022) — Automated CI/CD pipelines with Docker and GitHub Actions to containerize and deploy microservices to Heroku, minimizing environment drift; built data-cleaning workflows turning unstructured campus/event data into standardized, AI-ready datasets; ran system-integration testing across git workflows for stable releases.

Document Reconstruction Engine — Genetic Algorithm (Mar 2020) — Designed a structural reconstruction system using genetic algorithms to solve shredded-document reordering; compared three crossover models (Order, PMX, Alternative Position) in Python (Pandas) and Java, with quantitative analysis synthesized into a technical whitepaper.

EDUCATION

Brock University — Honours B.Sc. Computer Science, Minor in Economics, Jan 2019 – Jan 2023. Dean's Honour List (years 1, 2, and 3).

EXTRACURRICULAR & LEADERSHIP

- **Computer Science Club President (2020–2021)** — led coordination between club executives and external stakeholders to plan, scope, and deploy hybrid workshops; managed end-to-end event lifecycles and scaled a recurring workshop framework delivering technical training to a diverse student body.
- **International Collegiate Programming Contest (2021)** — analyzed complex logic problems in a high-pressure team setting, designing optimized algorithmic solutions under strict runtime and data constraints.

ADDITIONAL EXPERIENCE

- **Drop-in Tutor, Brock University (2022–2023)** — translated complex CS concepts into accessible lessons for a diverse student body; designed and delivered a structured C++ curriculum covering foundational object-oriented programming and system logic.
- **Crew / Team Lead, McDonald's (2019–2023)** — led kitchen operations and cross-functional teams under high-pressure conditions; awarded Employee of the Year.

VOLUNTEER WORK

- **Church Volunteer, St. Catharines (2019–present)** — run graphics, lighting, and audio for weekly services; assist with setup and teardown of the sanctuary; lead a weekly young-adults scripture group.
- **Canada Games 2022 Niagara (Aug 2022)** — provided tech support resolving technical issues during events, and served as a food provisioner coordinating meals for volunteers.

PORTFOLIO & LINKS

Personal Website |GitHub |LinkedIn |Topline